

ASHUTOSH SONAR

DEVOPS ENGINEER | CLOUD PLATFORM ENGINEER

Melbourne, FL | +1 (321) 389-9167 | ashusonar1998@gmail.com | [LinkedIn](#) | [GitHub](#)

PROFESSIONAL SUMMARY

DevOps Engineer with 4+ years of experience designing and scaling AWS-native infrastructure, CI/CD pipelines, and secure cloud platforms in regulated environments. Proven track record of improving deployment speed, system reliability, and cloud security posture using Infrastructure as Code, Kubernetes, and observability tooling. Experienced in building HIPAA-compliant, high-availability systems, enabling large-scale data and ML workloads. Strong expertise in cloud security, automation, and SRE practices, with a focus on performance optimization and zero-downtime delivery.

TECHNICAL SKILLS

Programming & Scripting: Python, Bash, Shell, Go

Cloud & Data Platforms: AWS (EC2, S3, Lambda, IAM, KMS, VPC, EKS, CloudWatch), Databricks

DevOps & CI/CD: Jenkins, ArgoCD, GitHub Actions, GitOps

Containerization & Infrastructure: Docker, Kubernetes (EKS), Terraform, Ansible

Monitoring & Observability: Prometheus, Grafana, ELK Stack

Security & Compliance: IAM, KMS, Secrets Management, HIPAA Compliance, Threat Modeling

Databases: PostgreSQL, MySQL, MongoDB, Cassandra, Neo4j, Redis

AI & Development Tools: Claude Code, OpenAI Codex, Cursor

Practices: SRE, IaC, Microservices, Zero-Downtime Deployment, Agile/Scrum

PROFESSIONAL EXPERIENCE

Florida Institute of Technology

Cloud Platform Engineer | Jan 2026 – Present

- Architected and deployed a petabyte-scale, multi-tenant healthcare data platform on AWS and Databricks, enabling secure access for 200+ researchers and reducing data delivery time from days to same-day.
- Designed and enforced HIPAA-compliant cloud architecture, implementing IAM least-privilege access, KMS encryption, VPC segmentation, and centralized audit logging, ensuring full regulatory compliance.
- Automated infrastructure provisioning using Terraform and Python, reducing environment setup time by 85% and enabling consistent deployments across dev, staging, and production.
- Implemented end-to-end observability (CloudWatch + ELK Stack), improving anomaly detection time by 70% and reducing incident response time by 30%.
- Optimized Databricks infrastructure to support 15–20 concurrent ML workloads, improving compute efficiency and reducing onboarding time for new teams by 60%.
- Built an Model Context Protocol (MCP) tool for the platform, enabling researchers to query medical imaging datasets, submit ML jobs, and manage workflows via natural language interfaces, eliminating dependency on direct API/SQL usage.
- Built an AI-assisted cloud data platform using Model Context Protocol (MCP), enabling researchers to query medical imaging datasets, submit ML jobs, and manage workflows via natural language interfaces, eliminating dependency on direct API/SQL usage.
- Designed and implemented 30+ provider methods across models, datasets, jobs, experiments, and governance layers, integrating Databricks for distributed compute and Unity Catalog for metadata management.
- Developed and executed a 200-test validation and benchmark suite covering MCP server contracts and provider surfaces, achieving 100% pass rate with sub-7-second execution time.
- Standardized heterogeneous medical imaging metadata (MRI, CT, X-ray) into unified schemas, enabling cross-modality dataset discovery and reproducible ML pipeline orchestration through containerized workflows.
- Collaborated with cross-functional teams to align platform capabilities with data science and clinical workflows, improving system usability and adoption.
- Strengthened platform reliability through proactive monitoring, alerting, and incident management aligned with SRE principles.

Graduate Research Assistant | Jul 2025 – Dec 2025

- Integrated automated threat modeling (STRIDE, PASTA, LINDDUN, DREAD) into CI/CD pipelines, preventing critical misconfigurations before production deployment.
- Enhanced pipeline security by embedding policy checks, validation gates, and compliance controls, improving deployment quality and reducing risk exposure.
- Analyzed system telemetry and logs to identify recurring reliability gaps, increasing early detection rates by 40% across multiple deployment cycles.

Coriolis Technology

DevOps Engineer (Member of Technical Staff) | Oct 2020 – Jul 2023

- Designed and managed CI/CD pipelines (Jenkins, ArgoCD) for 20+ microservices, reducing release cycles by 50% and increasing deployment frequency.
- Built and maintained EKS-based Kubernetes infrastructure, ensuring 99.999% uptime with zero-downtime deployment strategies and rolling updates.
- Automated cloud provisioning and deployment workflows using Python and Bash, cutting manual effort by 80% and improving release consistency.
- Implemented Infrastructure as Code (Terraform, Ansible) across multi-account AWS environments, eliminating configuration drift and improving scalability.
- Strengthened cloud security posture by integrating IAM policies, KMS encryption, and secrets management, reducing deployment-related security failures by 65%.
- Developed comprehensive monitoring solutions using Prometheus, Grafana, and CloudWatch, reducing MTTR by 40% and improving incident resolution speed.
- Improved system resilience through proactive alerting, logging, and performance tuning across distributed microservices architecture.
- Collaborated with engineering teams to implement DevOps best practices, improving development velocity and deployment reliability.

PROJECTS

Threat Model Recommender

- Developed a CLI tool to analyze cloud architectures and recommend hybrid threat modeling strategies, improving security coverage for AWS-based systems.

YAMGuard – CI/CD Configuration Validator

- Built a CI/CD validation tool to enforce YAML schema compliance and configuration integrity, reducing deployment failures and improving pipeline reliability.

EDUCATION

Florida Institute of Technology | **Aug 2023 – May 2025**

Master of Science in Computer Science

- **Thesis:** Integrating Threat Modeling (STRIDE, PASTA, DREAD) into CI/CD Pipelines

Savitribai Phule Pune University | **Aug 2016 – May 2020**

Bachelor of Engineering in Computer Engineering